

Nova Berman

novaberman@proton.me — (734) 263-4645 — github.com/aprzn123 — novaberman.com

Education

University of Michigan, College of Engineering 2022–2026
B.S.E. Computer Science — GPA 3.91/4.00

Courses: Operating Systems, Compilers, Parallel GPU Programming (CUDA), Automated Formal Verification, Cryptography, Programming Language Theory, Data Structures and Algorithms, Computer Organization, Fundamentals of Computer Science, Linear Algebra, Discrete Math, Electricity & Magnetism, Data Analysis, Robotics

Skills

Git, Java, Python, Rust, C++, C, Linux, JavaScript, HTML/CSS, SQL (SQLite), Make, CMake, Google Apps Script, L^AT_EX, (P)React, Docker

Work Experience

H3D, Inc. May 2025–Aug 2025, May 2026–Present
Software Intern

- Fixed numerous bugs and introduced numerous new features to post-processing software in C++.
- Researched, identified, and described a number of long-standing misunderstood bugs using Python.
- Proposed designs for upcoming features, solicited feedback from co-workers, and implemented them.
- Reimplemented previously non-production-ready C++ code for reliability, maintainability, and speed.
- Accelerated simulation algorithm performance from over 10 minutes to a few milliseconds.

UMich Department of Materials Science May 2023–April 2026
Research Assistant

- Designed Rust software to control motors in experimental setup using lasers to identify material properties.
- Built website with Flask that allows users to upload data to be processed by a machine learning model.
- Implemented and documented a Zola-based SSG setup for less-technical users to update a website.
- Collaborated with professor to modify software to ensure that it fulfills design requirements.

Lawrence Berkeley National Laboratory Aug 2022–Feb 2023
IT Intern, Cybersecurity Team

- Selected the best time-series database for the organization based on relevant use cases.
- Assembled Grafana dashboard to view performance data for any of approximately 100 servers.
- Adapted dashboard and database sources to target co-workers' needs when they arose.

Project Experience

Technology Advisor, Escher Cooperative House Fall 2024–Present

- Managed technological amenities for a house of 165 people.
- Implemented automations for house officers using Google Apps Script.
- Documented numerous systems used by house members.
- Responded to Requests for Assistance regarding house technology within a reasonable timeframe.
- Maintained and upgraded a wiki (MediaWiki) hosted on an AWS EC2 instance.

Individual Projects

- **Taskit** A TUI-based tool to track time spent working on tasks, with multiple active users. Enables users to categorize tasks, group categories of tasks together, and filter for a number of different properties. Built in Rust, using Ratatui and Inquire for the user interface. <https://github.com/aprzn123/taskit>
- **TheThirdCan** A browser extension to add features to one small forum website. Accepted contributions from community members. Used Javascript and HTML/CSS for the extension, and Rust Axum on the server side. github.com/aprzn123/TheThirdCan
- **RiPeN** A Reverse Polish Notation-based TUI calculator that integrates with Lua and Uiua to define custom operators, made with Rust and Ratatui. github.com/aprzn123/ripen

Hobbies/Interests: Writing game soundtracks, linguistics, productivity tools, server hosting, Linux audio